Pawan Kumar Thapaliya

PhD Candidate — Computational Neuroscience — Quantitative MRI & Deep Learning

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LinkedIn — Webpage — GitHub

Summary

Computational physicist and neurobiologist specializing in mechanistic modeling of astrocyte–neuron interactions, ionic signaling, and bioenergetics in stroke and epilepsy. Expertise in differential equation and Markov-based simulations, experimental data integration, and deep learning (CNNs, LSTMs, VAEs) applied to EEG and quantitative MRI. Experienced in teaching and mentoring undergraduate students in physics courses.

Education

Ph.D. Applied Physics (Computational Biophysics & Neurobiology) Aug 2019 – May 2026
University of South Florida, Tampa, FL

M.S. Physics (Computational Biophysics) Aug 2016 – Jul 2019
University of Texas Rio Grande Valley, TX

Technical Skills

Modeling & Math: ODEs, Markov models, ion channel & receptor kinetics, Monte Carlo
Machine Learning: PyTorch, TensorFlow, Keras, CNNs, LSTMs, VAEs
Programming: Python, MATLAB, R, SQL, FORTRAN, Bash
MRI & Medical Imaging: Quantitative MRI, diffusion MRI, **image segmentation, reconstruction, classification**
MRI / Neuro Libraries: PyTorch, **pydicom**, **nibabel**, MONAI, SimpleITK, MNE-Python
Scientific Stack: NumPy, SciPy, Pandas, Matplotlib, Seaborn, OpenCV
Tools: Git/GitHub, CUDA, Linux, Google Cloud Platform

Research & Industry Experience

PhD Researcher – Computational Neuroscience & Biophysics Aug 2019 – Present
University of South Florida

- Developed biophysical models revealing region-specific astrocytic Na^+/Ca^{2+} dynamics and ATP consumption.
- Modeled NMDA/AMPA receptor kinetics explaining neuronal energy demand differences.
- Quantified NBCe1 and NCX1 roles in ischemia-induced ionic overload using predictive simulations.
- Applied CNNs, LSTMs, and VAEs to EEG and imaging datasets for biomarker identification.

Quantitative MRI & Deep Learning Intern Jul 2025 – Present
Moffitt Cancer Center, Tampa, FL [Internship GitHub Repository]

- Conduct quantitative MRI using phantoms to optimize diffusion imaging protocols.
- Developed deep learning pipelines for MRI **reconstruction, segmentation, and classification**.
- Neuromatch Academy (2025): PCA & VAE training on MNIST/CIFAR-10 [Code]

Teaching Experience

- Adjunct Instructor and Graduate Teaching Assistant at University of South Florida and UTRGV, teaching Physics I & II courses to undergraduate students (150 students), including lectures, labs, grading, and mentoring.

Publications (Selected)

Thapaliya, P., et al. (2023). *Frontiers in Cellular Neuroscience* [Paper] [Code]
Everaerts, K., **Thapaliya, P.**, et al. (2023). *Cells* [Code]

Awards & Service

SfN Trainee Professional Development Award (2024)
USF Signature Doctoral Research Fellowship (2024–2026)
Reviewer: *Cellular Signaling*, *Journal of Theoretical Biology*